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Kinabalu Basin

Offshore NW Sabah, Malaysia — a geological dossier

Version 3 · October 2026 · Prepared for Arif Fazil
Public sources only. No proprietary well, seismic or company data.

What changed in v3. The regional section now runs from the Dangerous Grounds to East Sabah and the Sulu Sea (~830 km) and carries four competing working models of the tectonic driver, including a new, untested hypothesis: possible rift-axis intrusions along an inherited thinned axis beneath the Sabah Trough. Errors in v2 are corrected (Section 10).

1. What is the Kinabalu Basin?

In Malaysian petroleum usage the Kinabalu Basin is the offshore sedimentary basin along the north-west Sabah margin, from the waters off Labuan north toward the Malaysia–Philippines boundary. It is one of three producing basins in Malaysia, with the Malay and Sarawak basins (PETRONAS MPM 2025). It spans the NW Sabah shelf (<100 m water) and the deepwater fold-thrust belt (>1,000 m). Its outboard, north-western limit is the **Sabah Trough** (~2.9 km deep), beyond which lie the Layang-Layang Basin and the Dangerous Grounds.

Fields include Kinabalu (~500 MMstb oil in place; Bait 2003), Samarang, Kebabangan, Sumandak and Malikai. Recent wells have found oil beneath gas in deeper intervals (Ab Ghani et al. 2026).

2. Tectonic evolution

Sabah sits where the South China Sea, Sulu Sea and Celebes Sea basins meet. Its Cenozoic history runs from subduction, through collision, to post-subduction uplift and gravity-driven deformation.

Age	Event	Record
Eocene–Oligocene	Proto-South China Sea subducts SE beneath NW Borneo	Rajang–Crocker deep-marine turbidites accreted into the prism that forms the Crocker Range (Hutchison 2005)
~37 Ma	Sarawak Orogeny	Uplift and erosion of the Rajang Group (Hutchison 1996)
~32–16 Ma	South China Sea spreading	Dangerous Grounds blocks rifted and drifted south; carbonate platforms on block crests (Briais et al. 1993; Li et al. 2014)
~23–20 Ma	Sabah deformation · BMU	Uplift, erosion, mélange; deep-water to shallower sedimentation (Balaguru & Hall 2009)
~20–15 Ma	Sulu Sea back-arc opening	Extension and subsidence in East Sabah (Rangin & Silver 1991)
~16–15.5 Ma	End of spreading (RU/MMU) · DRU	Platforms drown in the Dangerous Grounds; Deep Regional Unconformity offshore Sabah (Hazebroek & Tan 1993)
Middle Miocene	North Sabah–Pagasa Wedge loaded; mud canopy	Mini-basins and ~1,900 km² of mud canopy (Morley et al. 2023)
~9.5–8.5 Ma	SRU · Late Miocene inversion	Shallow Regional Unconformity; published ages vary with biostratigraphic calibration
7.85–7.22 Ma	Kinabalu granite	U-Pb zircon (Cottam et al. 2010); rapid exhumation afterwards (Cottam et al. 2013)
Late Miocene–present	Delta progradation, toe-thrusts, inversion	Stage IV deltas; deepwater fold-thrust belt; 2015 Mw 6.0 Ranau earthquake shows ongoing deformation

What drives the deformation is disputed. Section 8 sets out four working models.

3. Stratigraphy

The producing interval is the post-DRU Stage IV succession, subdivided into IVA–IVG (Levell 1987). On the shelf it is a coastal to shallow-marine clastic system thickening into growth faults; outboard it passes into deepwater turbidites and mud, folded above an overpressured shale. Stage IVC records coastal progradation with mouth-bar and channel sands; IVD stacked coarsening-upward shoal complexes; IVE gradual transgression; IVF widespread transgression with laterally extensive sand sheets; IVG renewed shallowing (Bait 2003).

4. Kinabalu Field

Discovered in 1989 by KN-1, 55 km WNW of Labuan in ~54 m of water; oil in place ~500 MMstb; first oil December 1997, peak ~48,000 bbl/d. The trap is a dip closure against the SW–NE Kinabalu growth fault (8° WNW dip; spill to the NE). Main, Deep and East accumulations; best reservoir facies are poorly stratified sandstones (23% porosity, ~630 mD). A water-injection scheme from shallow aquifer sands raised recovery from ~21% to ~43% (Bait 2003).

5. Petroleum system

Element	Summary
Source	Not penetrated. Type III and II/III deltaic organic matter inferred from migrated hydrocarbons; generation from Late Miocene–Pliocene to present.
Reservoir	Post-DRU Stage IV coastal to shallow-marine sandstones (porosity 20–30%); deepwater turbidites outboard; Oligo–Miocene carbonates in the neighbouring Layang-Layang Basin (Tepat-1; Choi 2026).
Seal	Intraformational and transgressive marine shales (IVF, IVG).
Trap	Growth-fault dip closures on the shelf; thrust anticlines outboard; stratigraphic traps under-explored.
Key risk	Trap timing versus charge; Late Miocene–Pliocene inversion can breach or re-migrate.

6. Recent discoveries

Deeper oil beneath gas at Well A (2025), a deeper pool at Zoisit Deep-1, and reprocessed-seismic targets at Well B (Ab Ghani et al. 2026) have shifted the basin from gas-dominant to oil-prone. In the Layang-Layang Basin next door, carbonate discoveries (Tepat-1, Megah-1) prove an Oligo–Miocene carbonate play (Choi 2026). Receiver functions image a slab remnant 45–55 km beneath Sabah (Cornwell et al. 2025).

7. Regional section: Dangerous Grounds → East Sabah

An ~830-km dog-leg section from the Dangerous Grounds (NW) across the Sabah Trough, the outboard fold-thrust belt, the Kinabalu shelf, the Crocker Range and Mt Kinabalu, Central Sabah and the Sandakan Basin to the Sulu Sea (SE). The upper 15 km is drawn at ~17× vertical exaggeration and the lithosphere at ~3×. Shallow geometry is shared by all models; the region beneath the DRU and everything below 15 km depend on the model. All bodies are schematic interpretations, not seismic picks. Interactive version: arif-fazil.com/earth/kinabalu-cross-section.html.

Provenance — sources behind this dossier

This dossier is built from peer-reviewed and publicly released sources only. Key references by section:

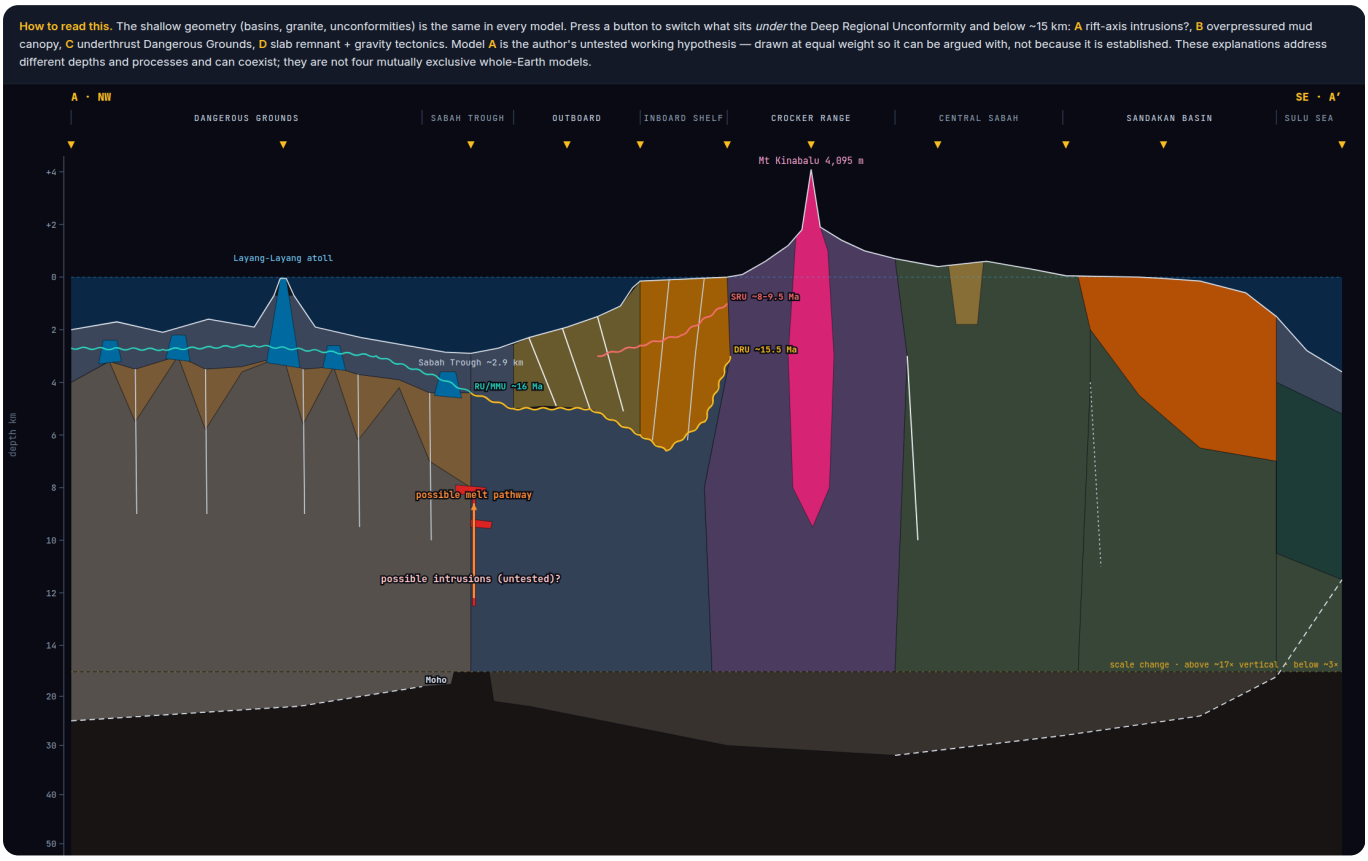
- **Section 2 (Tectonic evolution):** Balaguru & Hall 2009; Hutchison 2005; Briais et al. 1993; Rangin & Silver 1991.
- **Section 4 (Kinabalu field):** Bait 2003 (GSM Bulletin 47).
- **Section 6 (Recent discoveries):** Ab Ghani et al. EAGE 2026; Choi et al. EAGE 2026; Cornwell et al. JGR 2025.
- **Section 8 (Model A — rift-axis intrusions?):** Author's untested hypothesis; no public support yet.
- **Section 8 (Model B — mud canopy):** Morley et al. 2023, *Geosphere* 19(1): 291–326.
- **Section 8 (Model C — underthrust):** Hinz et al. 1989; Hazebroek & Tan 1993; Hutchison 2005.
- **Section 8 (Model D — slab + gravity):** Hall 2013; Cottam et al. 2010, 2013; King et al. 2010; Cornwell et al. 2025.
- **Background:** PETRONAS MPM 2025; Geological Survey of Malaysia Memoir 19.

No proprietary well, seismic or company data. No picks. No internal chronostratigraphy. Public sources only. Interpretive schematic hand-plotted from published values - not machine-derived from survey data or a canonical transect pipeline.

8. Four working models

A · rift-axis intrusions? — Failed-rift axis + possible transtensional intrusions UNTESTED WORKING HYPOTHESIS

Working hypothesis · Arif Fazil · 2026



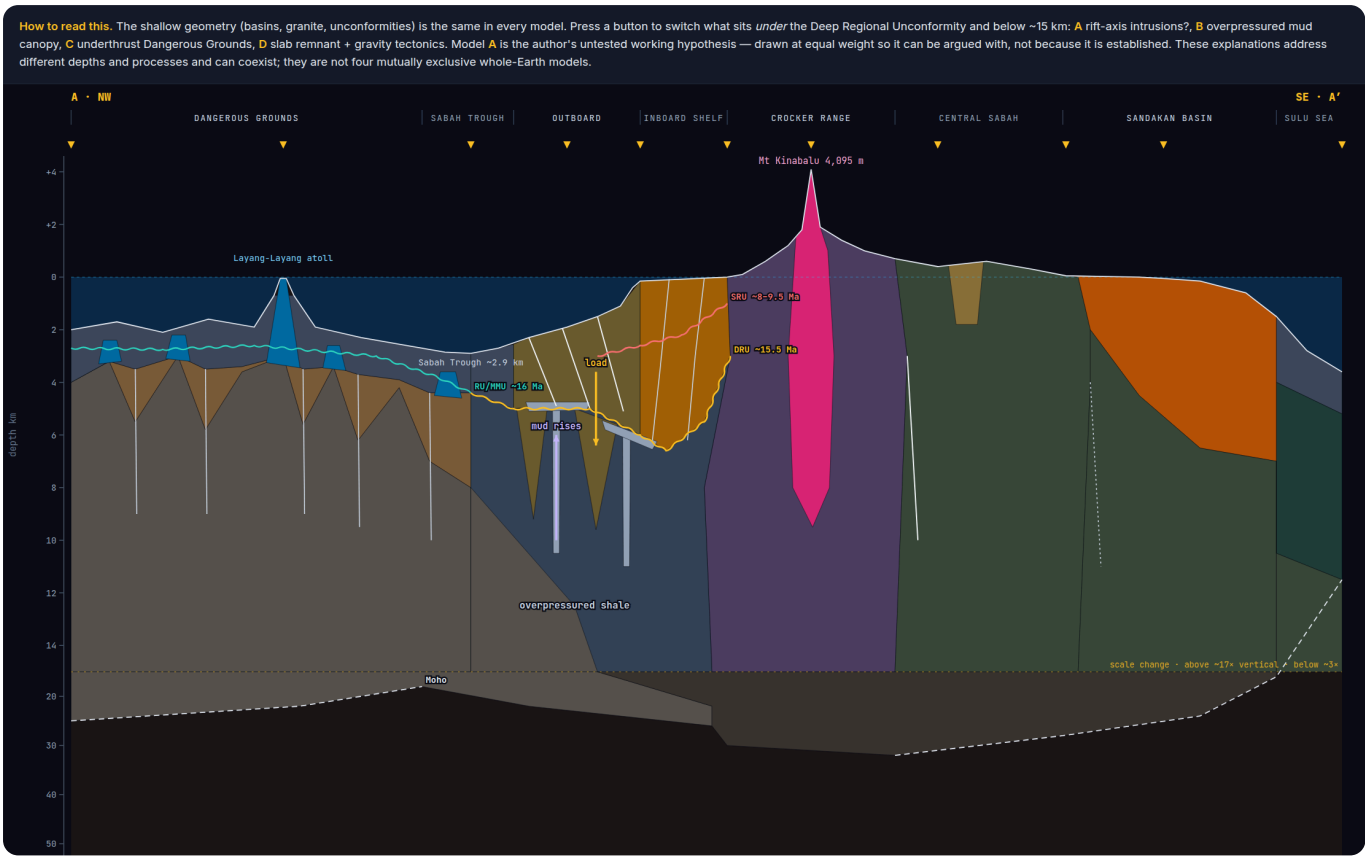
Section A–A′ under A · rift-axis intrusions?.

An inherited zone of thinned crust runs along the Sabah Trough; potential-field work suggests crustal thinning beneath the trough axis predating basin fill. If that axis carried any magmatism, it would most plausibly be sparse sills and dykes fed by transtensional reactivation of inherited faults — not a spreading ridge. Whether such intrusions exist is untested; the deep oil-beneath-gas maturation question is the practical reason to look.

MECHANISM	No plate-driving push is claimed. Candidate mechanism: localised melting or fluid migration along reactivated rift faults and transfer zones; slab-window or flexural-hinge sources remain possible but unproven.
TIMING	Not yet constrained. Two windows to test: (1) Oligocene–Early Miocene, at the end of rifting; (2) Middle–Late Miocene, coeval with DRU–SRU deformation.
STRENGTHS	Would provide a local heat source relevant to the deep oil beneath gas; would tie any dated intrusions along the trough into one testable body class. It is not offered as an explanation of the regional deformation budget.
PROBLEMS	No published magnetic lineation, dated basalt or heat-flow high on the trough axis. Published potential-field work reports a gravity low along the axis with inferred crustal thinning; any intrusion configuration must fit that field with a proper forward model, not a slab approximation. High-velocity bodies offshore are non-unique (carbonate or older Crocker blocks are alternatives). Detectability of small sills at regional-grid resolution has not yet been assessed.
KILL TEST	**Reject only a parameterised intrusion configuration whose required bodies are excluded by adequately resolved seismic and potential-field coverage; insufficient coverage yields INCONCLUSIVE. Finding no intrusions would not by itself prove burial-only maturation; finding intrusions would not by itself prove a slab-window origin.**

Model B · Morley et al. 2023 — Miocene mud canopy (NSPW) PUBLISHED

Morley et al. (2023) · Geosphere 19(1): 291 · 3D seismic + wells



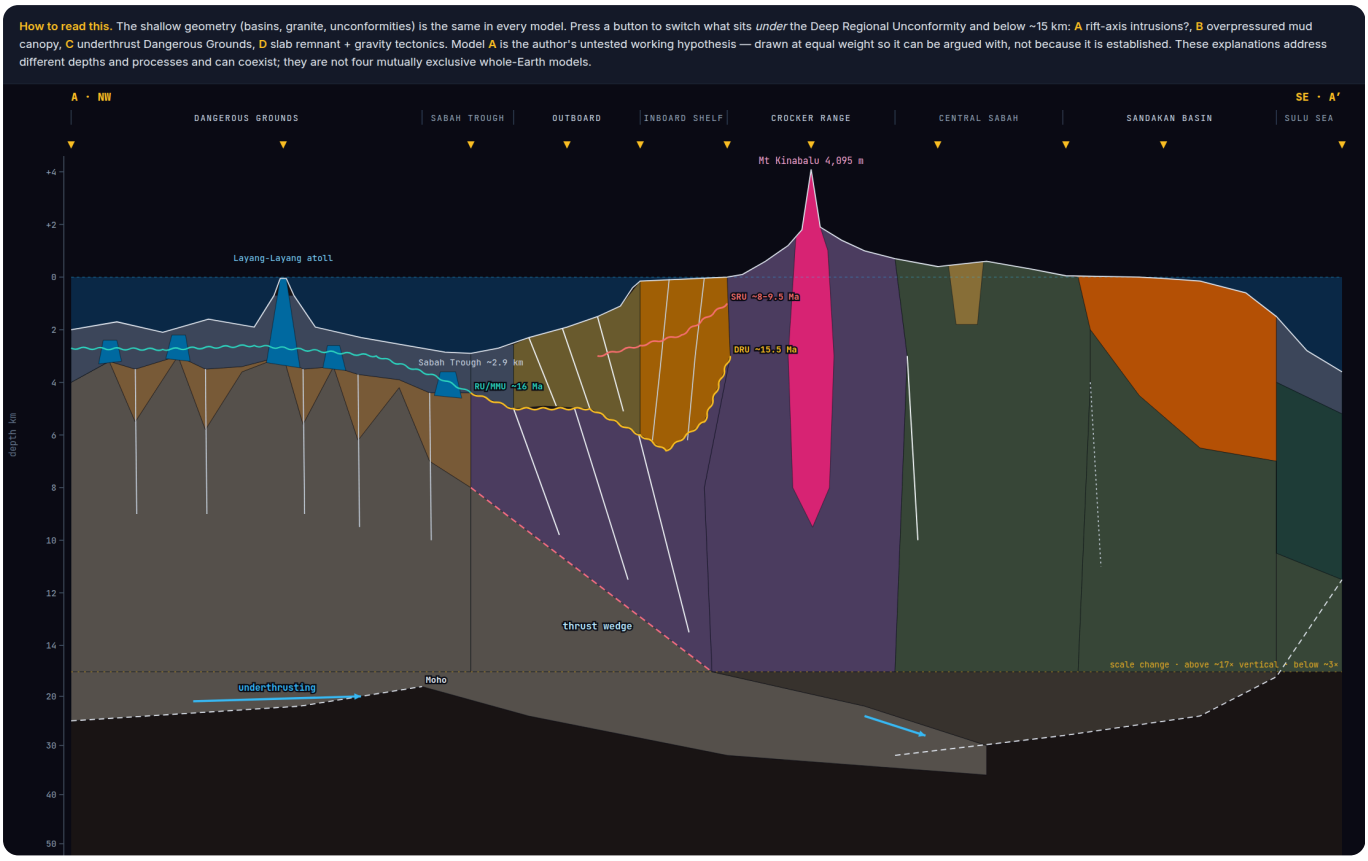
Section A–A′ under Model B · Morley et al. 2023.

The North Sabah–Pagasa Wedge is not mainly an accretionary wedge. After collision it was a body of overpressured shale, loaded by up to ~4 km of clastics in mini-basins. A short burst of mud volcanism then built a submarine mud canopy: about 50 mud-volcano centres, each 4–80 km², with ~1,900 km² of dense canopy — the largest and most complex such system described.

MECHANISM	Differential sedimentary loading of overpressured shale → shale mobility, mud pipes, chambers and volcanoes.
TIMING	Middle Miocene, after collision.
STRENGTHS	Built on 3D seismic and wells. Explains chaotic seismic facies, mini-basins and vent complexes in the outboard without invoking stacked thrust sheets.
PROBLEMS	Explains the wedge's internal structure, not the plate-scale driver: it still needs one of A, C or D underneath. Canopy level must be reconciled with the DRU age.
KILL TEST	**Wells through 'canopy' facies that prove in-situ accreted strata rather than extruded mud → reject for that area.**

Model C · classic — Continental underthrusting CLASSIC

Hinz et al. (1989); Hazebroek & Tan (1993); Hutchison (2005)



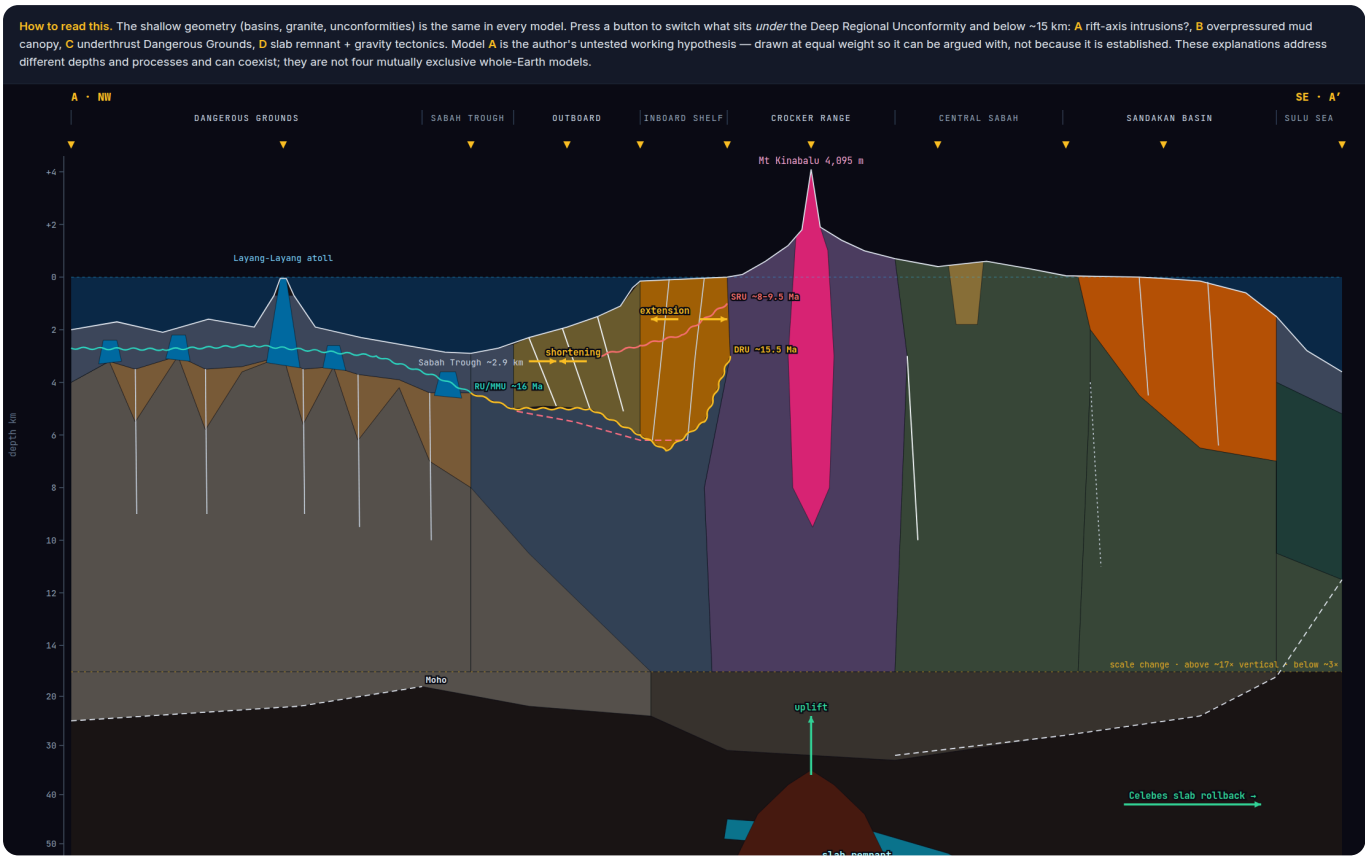
Section A–A′ under Model C · classic.

As the proto-South China Sea closed, Dangerous Grounds continental crust was carried south-east beneath NW Borneo. The Sabah Trough is the flexural foredeep, or remnant trench, on the down-going plate; the North Sabah–Pagasa Wedge is an accretionary thrust wedge; the DRU records the collision.

MECHANISM	Plate convergence and crustal underthrusting; flexural loading of the lower plate.
TIMING	Early to Middle Miocene collision; flexural loading continues.
STRENGTHS	Explains the deep, narrow trough, its flexural normal faults, the imbricate thrusts on older 2D seismic and the regional DRU.
PROBLEMS	Spreading ended ~16 Ma, yet shortening continues to the present; much young shortening is linked to delta loading. 3D seismic shows mobile shale where this model draws thrust sheets (B).
KILL TEST	**Deep seismic or receiver functions show no Dangerous Grounds crust beneath the shelf → reject.**

Model D · post-subduction — Slab remnant + gravity tectonics PUBLISHED

Hall (2013); Cottam et al. (2010, 2013); King et al. (2010); Cornwell et al. (2025)



Section A–A′ under Model D · post-subduction.

Subduction stopped in the Early Miocene, but a detached slab remnant still sits 45–55 km beneath Sabah. Post-subduction mantle flow and rollback of the Celebes Sea slab to the south-east drive uplift (Crocker Range, Kinabalu granite) and extension. The eroding mountains feed deltas whose weight drives shelf extension linked, through a shale detachment, to outboard toe-thrusts.

MECHANISM	Vertical mantle forces + rollback-driven extension + gravitational collapse of the delta wedge.
TIMING	Early Miocene end of subduction; Late Miocene to present uplift, granite and gravity tectonics.
STRENGTHS	Explains the slab image at 45–55 km, granite timing and fast exhumation, and the kinematic link between growth faults and toe-thrusts.
PROBLEMS	Gravity alone may not balance all outboard shortening (King et al. 2010 separate plate-scale and gravitational parts); the timing of Celebes Sea rollback is debated.
KILL TEST	**Outboard shortening far exceeds shelf extension with no detachment linkage → a plate-scale push (A or C) is needed.**

9. What would decide it

Observation	A	B	C	D	Status
Linear magnetic + gravity high along the trough axis (Tepat-1 strike line)	✓	○	✗	○	open
Dated basalt / dolerite in trough-flank wells	✓	○	○	○	open
Sills, saucer intrusions, hydrothermal vents in trough fill (3D seismic)	✓	○	✗	○	open
Heat flow / maturity peaks on the trough axis	✓	○	✗	○	open
3D-seismic pipes/chambers/mini-basins at Middle Miocene level (interpreted as mobile shale)	○	✓	✗	○	observed Morley et al. 2023
Outboard shortening exceeds shelf extension	✓	○	✓	✗	open contested; King et al. 2010
West-dipping discontinuity at 45–55 km (interpreted as detached slab remnant)	○	○	✓	✓	observed Cornwell et al. 2025
Granite 7.85–7.22 Ma with rapid exhumation	○	○	○	✓	observed Cottam et al. 2010, 2013

✓ predicted · ✗ contradicted or not expected · ○ silent. The four observations that would test Model A exist in industry data but are not public. Model A is drawn at the same weight as the published models so it can be argued with, not because it is established.

10. Corrections from v2 (July 2026)

- Kinabalu granite: v2 gave ~10–8 Ma. The U-Pb age is **7.85–7.22 Ma** (Cottam et al. 2010). The unsourced "~1,000 km² at depth" is removed.
- Sabah Trough: v2 called it the basin's eastern boundary. It is the **outboard, north-western** limit.
- SRU: v2 gave a single 8.6 Ma. Published ages range ~8.5–9.5 Ma depending on biostratigraphic calibration; it is shown as a range.
- The regional section is extended from ~600 km to ~830 km to reach the Sulu Sea, and now carries four models instead of one picture.

11. References

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